

DennisJCarroll.com — Gatsby Site Fix Implementation Guide

A Gatsby-specific, step-by-step roadmap organized by priority. Copy-paste where possible, adapt paths to your actual file structure.

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P0 — Immediate Fixes (10 minutes)

1. Fix “Contact Me” Button

File: `src/pages/index.js` (or `src/sections/Hero.js`)

Problem: The “Contact Me” button does nothing.

Fix — Option A: Smooth scroll to footer contact form

```
// In your Hero section, find the "Contact Me" button and update:

<a
  href="#contact"
  onClick={(e) => {
    e.preventDefault();
    document.getElementById('contact')?.scrollIntoView({
      behavior: 'smooth',
      block: 'start',
    });
  }}
  className="...your-existing-classes..."
>
  Contact Me
</a>
```

Then verify your contact section has `id="contact"`:

```
// In your footer or contact section wrapper
<section id="contact" className="...">
  {/* contact form content */}
</section>
```

Fix — Option B: Navigate to `/about/#contact`

```
import { Link } from 'gatsby';
```

```
<Link
  to="/about/#contact"
  className="..."
>
  Contact Me
</Link>
```

2. Fix Footer Dead Links

File: `src/components/Footer.js` (or wherever footer quick links are defined)

Problem: “Development Projects” and “Interactive Apps” links point to non-existent pages.

Current (likely):

```
<a href="/development-projects/">Development Projects</a>
<a href="/interactive-apps/">Interactive Apps</a>
```

Fix: Point both to the unified `/apps/` page, or remove them:

```
// Option A: Redirect to /apps/
[[
  { label: 'Home', href: '/' },
  { label: 'Apps & Projects', href: '/apps/' }, // keep this
  // REMOVE: { label: 'Development Projects', href: '/development-projects/' },
  // REMOVE: { label: 'Interactive Apps', href: '/interactive-apps/' },
  { label: 'Creative', href: '/stories/' },
  { label: 'About', href: '/about/' },
  { label: 'Contact', href: '/about/#contact' },
].map(link => (
  <a key={link.href} href={link.href} className="...">
    {link.label}
  </a>
))}]
```

3. Fix “The Genesis Protocols” Date

File: `src/pages/stories.js` or `src/data/stories.js` (wherever story metadata is defined)

Problem: Date shows April 05, 2026 (future date).

Fix: Change to actual publish date:

```
{
  title: "The Genesis Protocols (QSRI File 7734-C)",
  date: "2025-04-05", // or correct date
  // ...
}
```

P1 — Quick Wins (1–2 hours)

4. Simplify Theme Toggle

File: `src/components/ThemeToggle.js` (or `src/components/Layout.js`)

Problem: Four unlabeled symbols () — users don't understand them.

Fix: Replace with a single sun/moon toggle with tooltip label.

```
// ThemeToggle.js
import React, { useState, useEffect } from 'react';

const ThemeToggle = () => {
  const [theme, setTheme] = useState('dark');

  useEffect(() => {
    const saved = localStorage.getItem('theme') || 'dark';
    setTheme(saved);
    document.documentElement.setAttribute('data-theme', saved);
  }, []);

  const toggleTheme = () => {
    const next = theme === 'dark' ? 'light' : 'dark';
    setTheme(next);
    document.documentElement.setAttribute('data-theme', next);
    localStorage.setItem('theme', next);
  };

  return (
    <button
      onClick={toggleTheme}
      aria-label={`Switch to ${theme === 'dark' ? 'light' : 'dark'} theme`}
      title={` ${theme === 'dark' ? 'Light' : 'Dark'} mode`}
      className="theme-toggle-btn"
    >
      {theme === 'dark' ? (
        // Sun icon for "switch to light"
        <svg width="20" height="20" viewBox="0 0 24 24" fill="none" stroke="currentColor"
          ↪ strokeWidth="2">
          <circle cx="12" cy="12" r="5" />
          <path d="M12 1v2M12 21v2M4.22 4.22l1.42 1.42M18.36 18.36l1.42 1.42M1 12h2M21 12h2M4.22
            ↪ 19.78l1.42-1.42M18.36 5.64l1.42-1.42" />
        </svg>
      ) : (
        // Moon icon for "switch to dark"
        <svg width="20" height="20" viewBox="0 0 24 24" fill="none" stroke="currentColor"
          ↪ strokeWidth="2">
          <path d="M21 12.79A9 9 0 1 1 11.21 3 7 7 0 0 0 21 12.79z" />
        </svg>
      )}
    </button>
  );
};
```

```
};  
  
export default ThemeToggle;
```

If you want to keep 4 themes, add a labeled dropdown instead:

```
const THEMES = [  
  { id: 'dark', label: 'Dark', icon: '$$\vcenter{\hbox{\includegraphics[height=\ht\strutbox]{/usr/  
    ↪ share/docminer/emoji/1f319.png}}}}$$' },  
  { id: 'light', label: 'Light', icon: '$$\vcenter{\hbox{\includegraphics[height=\ht\strutbox]{/usr/  
    ↪ share/docminer/emoji/2600-fe0f.png}}}}$$' },  
  { id: 'midnight', label: 'Midnight', icon: '$$\vcenter{\hbox{\includegraphics[height=\ht\strutbox]  
    ↪ {/usr/share/docminer/emoji/2726.png}}}}$$' },  
  { id: 'ocean', label: 'Ocean', icon: '💠' },  
];  
  
// Use a dropdown with text labels instead of raw symbols
```

5. Replace “100% Passion” Stat

File: src/pages/about.js

Current:

```
const stats = [  
  { value: '15', label: 'Projects Completed', suffix: '+' },  
  { value: '5', label: 'Years Experience', suffix: '+' },  
  { value: '10', label: 'Technologies', suffix: '+' },  
  { value: '100', label: 'Passion', suffix: '%' }, // <-- REMOVE THIS  
];
```

Better options (pick one):

```
// Option A — Apps built (matches your headline count)  
{ value: '24', label: 'Interactive Apps Built', suffix: '+' },  
  
// Option B — GitHub metric (if you can fetch or hardcode)  
{ value: '1.2K', label: 'GitHub Commits', suffix: '+' },  
  
// Option C — Content volume  
{ value: '3', label: 'Creative Universes', suffix: '' },  
  
// Option D — Just remove the 4th stat, 3 is fine  
const stats = [  
  { value: '24', label: 'Interactive Apps Built', suffix: '+' },  
  { value: '5', label: 'Years Experience', suffix: '+' },  
  { value: '10', label: 'Technologies', suffix: '+' },  
];
```

6. Add “Back to Projects” Banner to Standalone Apps

File: Create `src/components/StandaloneAppShell.js`

Problem: Apps like `neural_theory_lab_v2.html` break from site branding.

Fix: Create a thin branded banner component that wraps standalone HTML apps.

```
// src/components/StandaloneAppShell.js
import React from 'react';
import { Link } from 'gatsby';

const StandaloneAppShell = ({ children, appTitle }) => (
  <div className="min-h-screen bg-[#0a0a0a]">
    {/* Branded top bar */}
    <div className="sticky top-0 z-50 bg-[#0f0f1a]/90 backdrop-blur-sm border-b border-cyan-900/30">
      <div className="max-w-7xl mx-auto px-4 h-12 flex items-center justify-between">
        <div className="flex items-center gap-4">
          <Link
            to="/apps/"
            className="text-cyan-400 hover:text-cyan-300 text-sm font-medium flex items-center gap-1
              ↪ transition-colors"
          >
            <svg width="16" height="16" viewBox="0 0 24 24" fill="none" stroke="currentColor"
              ↪ strokeWidth="2">
              <path d="M19 12H5M12 19l-7-7 7-7" />
            </svg>
            Back to Projects
          </Link>
          <span className="text-gray-600">|</span>
          <span className="text-gray-300 text-sm font-medium">{appTitle}</span>
        </div>
        <Link to="/" className="text-cyan-400 font-bold text-sm tracking-wide">
          DJC
        </Link>
      </div>
    </div>

    {/* App content iframe or embedded */}
    <main className="w-full">
      {children}
    </main>
  </div>
);

export default StandaloneAppShell;
```

Usage in a Gatsby page wrapper (e.g., `src/pages/apps/neural-theory-lab.js`):

```
import React from 'react';
import StandaloneAppShell from '../../../components/StandaloneAppShell';

const NeuralTheoryLabPage = () => (
```

```

<StandaloneAppShell appTitle="Neural Network Theory Lab v2">
  /* Option 1: Embed the HTML directly */
  <div
    dangerouslySetInnerHTML={{
      __html: require('!raw-loader!../../static/apps/neural_theory_lab_v2.html')
    }}
  />
  /* Option 2: Use an iframe */
  /* <iframe
    src="/apps/neural_theory_lab_v2.html"
    className="w-full h-screen border-0"
    title="Neural Theory Lab"
  /> */
</StandaloneAppShell>
);

export default NeuralTheoryLabPage;

```

Alternative — simpler: Add this `<style>` + `<div>` directly to each standalone `.html` file in your `/static/apps/` folder:

```

<!-- Add this at the top of each standalone HTML app -->
<style>
  .dj-c-app-bar {
    position: sticky;
    top: 0;
    z-index: 9999;
    background: rgba(15, 15, 26, 0.95);
    backdrop-filter: blur(8px);
    border-bottom: 1px solid rgba(34, 211, 238, 0.15);
    padding: 0 1rem;
    height: 44px;
    display: flex;
    align-items: center;
    justify-content: space-between;
    font-family: -apple-system, BlinkMacSystemFont, 'Segoe UI', sans-serif;
  }
  .dj-c-app-bar a { color: #22d3ee; text-decoration: none; font-size: 13px; font-weight: 500; }
  .dj-c-app-bar a:hover { color: #67e8f9; }
  .dj-c-app-bar .title { color: #d1d5db; font-size: 13px; }
  .dj-c-app-bar .logo { color: #22d3ee; font-weight: 700; letter-spacing: 0.05em; }
</style>

<div class="dj-c-app-bar">
  <div style="display:flex;align-items:center;gap:12px;">
    <a href="/apps/">- Back to Projects</a>
    <span style="color:#4b5563;">|</span>
    <span class="title">Neural Network Theory Lab v2</span>
  </div>
  <a href="/" class="logo">DJC</a>
</div>

```

P2 — Content & Structural (3–5 hours)

7. Rewrite About Page Bio

File: `src/pages/about.js` (or `src/sections/AboutBio.js`)

Current (generic):

```
I'm a passionate technologist with a deep interest in data science, project development, and creative expression. My background spans multiple disciplines...
```

Better (specific):

```
const bioParagraphs = [
  `I build interactive tools that make complex ideas intuitive. Over the past 5 years, I've created 24+
  ↪ standalone web applications spanning neural network visualizations, Bayesian analytics
  ↪ dashboards, real-time ML training environments, and mathematical exploration tools — all designed
  ↪ to run in the browser.`,
  `My current focus is at the intersection of deep learning and interpretability: understanding not just
  ↪ what neural networks learn, but how and why they learn it. Recent projects include an Agent Trace
  ↪ Viewer for SWE-agent interaction analysis, a Mechanistic Interpretability visualizer for
  ↪ transformer architectures, and an Edge AI Training Lab powered by TensorFlow.js.`,
  `I'm also the author of three original fictional universes — including Crack in the Veil, a
  ↪ post-humanity sci-fi saga, and A Chronicle of Lyos, a fantasy world where dead gods' bloodlines
  ↪ still remember.`,
  `When I'm not coding or writing, I'm exploring reinforcement learning, graph neural networks, and the
  ↪ mathematics of complex dynamical systems.`,
];

// In your component:
<section className="max-w-2xl mx-auto text-gray-300 leading-relaxed space-y-6">
  {bioParagraphs.map((p, i) => (
    <p key={i}>{p}</p>
  ))}
</section>
```

Also update the subtitle:

```
// From:
"Data Scientist • Full-Stack Developer • Creative Technologist"

// To (more specific):
"Interactive ML Tools • Bayesian Analytics • Creative Fiction"
```

8. Redesign Homepage “Featured Work” Section

File: `src/sections/FeaturedWork.js` (or `src/pages/index.js`)

Current: Shows 3 category cards (Apps & Projects, Stories & More, About Me).

Replace with: 3 actual flagship projects with thumbnails.

```
// src/sections/FeaturedWork.js
import React from 'react';
import { Link } from 'gatsby';
import { StaticImage } from 'gatsby-plugin-image';

const featuredProjects = [
  {
    title: 'Bayesian Analysis of Airbnb Seattle Market',
    description:
      'Enterprise-grade hierarchical Bayesian analytics platform with uncertainty quantification and an  

      ↪ interactive Streamlit dashboard for pricing optimization.',
    tags: ['Python', 'PyMC', 'Streamlit', 'Bayesian Statistics'],
    image: '../images/projects/bayesian-airbnb.jpg', // add screenshot
    links: [
      { label: 'GitHub', url: 'https://github.com/Dennis-J-Carroll/  

      ↪ bayesian-analysis-of-airbnb-seattle-market' },
      { label: 'Live Demo', url: 'https://bayesian-analysis-dashboard.streamlit.app/' },
    ],
    accent: 'cyan',
  },
  {
    title: 'Understanding Layer — Agent Trace Viewer',
    description:
      'Live visualization of real GPT-4.1 SWE-agent traces across 46 events. Renders the  

      ↪ InteractionTrace schema with causal links, fabrication detection, and annotation callouts.',
    tags: ['JavaScript', 'AI Observability', 'LLM'],
    image: '../images/projects/agent-trace.jpg',
    links: [
      { label: 'Launch App', url: '/apps/understanding-layer-demo.html' },
    ],
    accent: 'purple',
    isNew: true,
  },
  {
    title: 'Neural Network Theory Lab v2',
    description:
      'Enhanced interactive laboratory for exploring neural network architectures, training dynamics,  

      ↪ and theoretical foundations with LaTeX math rendering and attention heat maps.',
    tags: ['TensorFlow.js', 'React', 'D3.js'],
    image: '../images/projects/neural-lab.jpg',
    links: [
      { label: 'Launch App', url: '/apps/neural_theory_lab_v2.html' },
    ],
    accent: 'emerald',
    isNew: true,
  },
];
```



```

        className="text-xs bg-gray-800 text-cyan-400/80 px-2 py-1 rounded"
      >
        {tag}
      </span>
    )))
  </div>

  { /* Links */ }
  <div className="flex gap-4">
    {project.links.map((link) => (
      <a
        key={link.label}
        href={link.url}
        target={link.url.startsWith('http') ? '_blank' : undefined}
        rel={link.url.startsWith('http') ? 'noopener noreferrer' : undefined}
        className="text-sm text-cyan-400 hover:text-cyan-300 font-medium inline-flex
          ↪ items-center gap-1"
      >
        {link.label}
        {link.url.startsWith('http') && (
          <svg width="12" height="12" viewBox="0 0 24 24" fill="none" stroke="currentColor"
            ↪ strokeWidth="2">
            <path d="M18 13v6a2 2 0 0 1-2 2H5a2 2 0 0 1-2 2V8a2 2 0 0 1 2-2h6M15 3h6v6M10 14L21
              ↪ 3" />
            </svg>
          )}
        </a>
      )))
    </div>
  </div>
  </article>
  )))
</div>

<div className="text-center mt-12">
  <Link
    to="/apps/"
    className="inline-flex items-center gap-2 text-cyan-400 hover:text-cyan-300 font-medium"
  >
    View all 24+ projects <span>↪</span>
  </Link>
</div>
</section>
);

export default FeaturedWork;

```

9. Add Project Screenshots to App Cards

File: src/pages/apps.js

Two approaches — pick based on your setup:

Approach A: Auto-generate placeholders (quick)

Add a visual color-hash pattern or gradient based on project name so each card has unique visual identity:

```
// Helper: generate a deterministic gradient from project title
const getProjectGradient = (title) => {
  const gradients = [
    'from-cyan-900/40 to-blue-900/40',
    'from-purple-900/40 to-pink-900/40',
    'from-emerald-900/40 to-teal-900/40',
    'from-amber-900/40 to-orange-900/40',
    'from-rose-900/40 to-red-900/40',
    'from-indigo-900/40 to-violet-900/40',
  ];
  const hash = title.split('').reduce((a, c) => a + c.charCodeAt(0), 0);
  return gradients[hash % gradients.length];
};

// In your project card:
<div className={`aspect-video rounded-t-xl bg-gradient-to-br ${getProjectGradient(project.title)} flex
  ↪  items-center justify-center`}>
  {/* App icon or initials */}
  <span className="text-3xl font-bold text-white/20">
    {project.title.split(' ').map(w => w[0]).join('').slice(0, 3)}
  </span>
</div>
```

Approach B: Manual screenshots (better, more effort)

1. Capture screenshots of each app (use browser devtools or a tool like `pageres-cli`)
2. Save to `src/images/projects/`
3. Use `gatsby-plugin-image`:

```
import { StaticImage } from 'gatsby-plugin-image';

// In your card:
<div className="aspect-video rounded-t-xl overflow-hidden bg-gray-800">
  <StaticImage
    src="../../../images/projects/neural-lab.jpg"
    alt={project.title}
    className="w-full h-full object-cover"
    placeholder="blurred"
  />
</div>
```

Recommended: Start with Approach A (10 minutes), then gradually add real screenshots over time.

10. Add “What I’m Learning Now” Section

File: src/pages/about.js

Add after the bio:

```
<section className="mt-20">
  <h3 className="text-xl font-bold text-white mb-6 text-center">
    Currently Exploring
  </h3>

  <div className="grid md:grid-cols-3 gap-4 max-w-4xl mx-auto">
    {[
      {
        topic: 'Mechanistic Interpretability',
        detail: 'Understanding how transformer attention heads and MLP blocks represent concepts',
        link: '/apps/mech-interp-viz.html',
      },
      {
        topic: 'Graph Neural Networks',
        detail: 'Extending neural architectures to non-Euclidean graph-structured data',
        link: null,
      },
      {
        topic: 'Reinforcement Learning Theory',
        detail: 'Connecting RL optimization to dynamical systems and chaos theory',
        link: '/apps/chaos-error-rl.html',
      },
    ]}.map((item) => (
      <div
        key={item.topic}
        className="bg-gray-900/50 border border-gray-800 rounded-lg p-5 hover:border-cyan-500/30
          ↪ transition-colors"
      >
        <h4 className="text-cyan-400 font-semibold text-sm mb-2">
          {item.topic}
        </h4>
        <p className="text-gray-400 text-sm leading-relaxed mb-3">
          {item.detail}
        </p>
        {item.link && (
          <a
            href={item.link}
            className="text-xs text-cyan-500 hover:text-cyan-400 font-medium"
          >
            See experiment 
          </a>
        )}
      </div>
    ))}
    </div>
  </section>
```

11. Improve Hero Section

File: src/sections/Hero.js

Option A: Add a personal element

```
// Below the tagline, add a compact "what I do" row:
<div className="mt-8 flex flex-wrap justify-center gap-6 text-sm text-gray-400">
  {[
    { icon: '$$\vcenter{\hbox{\includegraphics[height=\ht\strutbox]{/usr/share/docminer/emoji/
    ↪ 1f9e0.png}}}}$$', label: 'ML & Data Science' },
    { icon: '$$\vcenter{\hbox{\includegraphics[height=\ht\strutbox]{/usr/share/docminer/emoji/
    ↪ 26a1.png}}}}$$', label: 'Interactive Web Apps' },
    { icon: '$$\vcenter{\hbox{\includegraphics[height=\ht\strutbox]{/usr/share/docminer/emoji/
    ↪ 270d-fe0f.png}}}}$$', label: 'Creative Fiction' },
  ].map(item => (
    <span key={item.label} className="flex items-center gap-2">
      <span>{item.icon}</span>
      <span>{item.label}</span>
    </span>
  ))}
</div>

// Remove emojis if they don't fit your aesthetic — use inline SVGs instead:
<svg className="w-4 h-4 text-cyan-400">...</svg>
```

Option B: Tighten spacing and add scroll indicator

```
// After CTA buttons, add:
<div className="mt-12 animate-bounce">
  <svg
    width="20" height="20" viewBox="0 0 24 24"
    fill="none" stroke="currentColor" strokeWidth="2"
    className="text-gray-600 mx-auto"
  >
    <path d="M12 5v14M19 12l-7 7-7-7" />
  </svg>
</div>
```

P3 — Feature Additions (1–2 days)

12. Add Search & Filter to Apps Page

File: src/pages/apps.js

```
// At the top of the component:
const [searchQuery, setSearchQuery] = useState('');
const [activeFilter, setActiveFilter] = useState('All');

const categories = ['All', 'AI/ML', 'Data Science', 'Math', 'Tools', 'Creative', 'Education'];
```

```

const filteredProjects = useMemo(() => {
  return projects.filter(p => {
    const matchesSearch = !searchQuery ||
      p.title.toLowerCase().includes(searchQuery.toLowerCase()) ||
      p.description.toLowerCase().includes(searchQuery.toLowerCase()) ||
      p.tags.some(t => t.toLowerCase().includes(searchQuery.toLowerCase()));

    const matchesCategory = activeFilter === 'All' || p.category === activeFilter;

    return matchesSearch && matchesCategory;
  });
}, [projects, searchQuery, activeFilter]);

// Render:
<section className="max-w-7xl mx-auto px-4 py-8">
  {/* Search bar */}
  <div className="mb-6">
    <input
      type="text"
      placeholder="Search projects, tags, technologies..."
      value={searchQuery}
      onChange={(e) => setSearchQuery(e.target.value)}
      className="w-full max-w-md bg-gray-900 border border-gray-700 rounded-lg px-4 py-2.5 text-white
        ⇨ placeholder-gray-500 focus:outline-none focus:border-cyan-500 transition-colors"
    />
  </div>

  {/* Category pills */}
  <div className="flex flex-wrap gap-2 mb-8">
    {categories.map(cat => (
      <button
        key={cat}
        onClick={() => setActiveFilter(cat)}
        className={`px-4 py-1.5 rounded-full text-sm font-medium transition-colors ${
          activeFilter === cat
            ? 'bg-cyan-500 text-black'
            : 'bg-gray-800 text-gray-400 hover:bg-gray-700 hover:text-white'
        }`}
      >
        {cat}
      </button>
    ))}
  </div>

  {/* Results count */}
  <p className="text-gray-500 text-sm mb-6">
    {filteredProjects.length} project{filteredProjects.length !== 1 ? 's' : ''} found
  </p>

  {/* Grid */}

```

```

<div className="grid md:grid-cols-2 lg:grid-cols-3 gap-6">
  {filteredProjects.map(project => (
    <ProjectCard key={project.title} {...project} />
  ))}
</div>

{/* Empty state */}
{filteredProjects.length === 0 && (
  <div className="text-center py-20 text-gray-500">
    No projects match your search.
    <button
      onClick={() => { setSearchQuery(''); setActiveFilter('All'); }}
      className="ml-2 text-cyan-400 hover:underline"
    >
      Clear filters
    </button>
  </div>
)}
</section>

```

13. Add Resume Download / “Available For Work” Badge

File: src/pages/about.js

Add a CTA section:

```

<section className="mt-16 text-center">
  <div className="inline-flex items-center gap-3 bg-gray-900 border border-gray-800 rounded-full px-6
    ↵ py-3">
    <span className="relative flex h-3 w-3">
      <span className="animate-ping absolute inline-flex h-full w-full rounded-full bg-emerald-400
        ↵ opacity-75" />
      <span className="relative inline-flex rounded-full h-3 w-3 bg-emerald-500" />
    </span>
    <span className="text-gray-300 text-sm">
      Open to collaboration and interesting projects
    </span>
  </div>

  <div className="mt-6 flex justify-center gap-4">
    <a
      href="/Dennis-Carroll-Resume.pdf"
      download
      className="bg-cyan-500 hover:bg-cyan-400 text-black font-semibold px-6 py-2.5 rounded-lg text-sm
        ↵ transition-colors inline-flex items-center gap-2"
    >
      <svg width="16" height="16" viewBox="0 0 24 24" fill="none" stroke="currentColor" strokeWidth="2">
        <path d="M21 15v4a2 2 0 0 1-2 2H5a2 2 0 0 1-2-2v-4M7 10l5 5 5-5M12 15V3" />
      </svg>
      Download Resume
    </a>
  </div>
</section>

```

```

</a>
<a
  href="mailto:your@email.com"
  className="border border-gray-700 hover:border-cyan-500 text-gray-300 hover:text-white px-6
    ↪ py-2.5 rounded-lg text-sm transition-colors"
>
  Get in Touch
</a>
</div>
</section>

```

14. Particle Animation Performance

File: src/components/ParticleBackground.js (or wherever canvas animation lives)

Pause animation when off-screen to save CPU/battery:

```

import { useEffect, useRef } from 'react';

const ParticleBackground = () => {
  const canvasRef = useRef(null);
  const animRef = useRef(null);
  const isVisibleRef = useRef(true);

  useEffect(() => {
    const canvas = canvasRef.current;
    if (!canvas) return;

    // Pause when tab is hidden or canvas is off-screen
    const observer = new IntersectionObserver(
      ([entry]) => {
        isVisibleRef.current = entry.isIntersecting;
      },
      { threshold: 0 }
    );
    observer.observe(canvas);

    const handleVisibility = () => {
      isVisibleRef.current = document.visibilityState === 'visible';
    };
    document.addEventListener('visibilitychange', handleVisibility);

    const ctx = canvas.getContext('2d');

    const animate = () => {
      if (isVisibleRef.current) {
        // your existing draw code
        ctx.clearRect(0, 0, canvas.width, canvas.height);
        // ... particles ...
      }
    }
  });

```



```

    animRef.current = requestAnimationFrame(animate);
  };

  animate();

  return () => {
    cancelAnimationFrame(animRef.current);
    observer.disconnect();
    document.removeEventListener('visibilitychange', handleVisibility);
  };
}, []);

return <canvas ref={canvasRef} className="fixed inset-0 pointer-events-none" />;
};

```

Also reduce particle count for mobile:

```
const PARTICLE_COUNT = window.innerWidth < 768 ? 15 : 40;
```

Project File Map

Use this to locate where changes should go. Adjust to match your actual structure.

```

dennisjcarroll.com/
├─ src/
│   ├─ components/
│   │   ├─ Layout.js           # wraps all pages, add theme toggle here
│   │   ├─ Navbar.js          # top navigation
│   │   ├─ Footer.js          # footer + quick links ← fix dead links here
│   │   ├─ ThemeToggle.js     # ← simplify or add labels
│   │   ├─ ParticleBackground.js # ← add IntersectionObserver
│   │   ├─ StandaloneAppShell.js # ← NEW: wrap standalone HTML apps
│   │   └─ ProjectCard.js     # ← add gradient thumbnail
│   ├─ sections/
│   │   ├─ Hero.js            # ← fix Contact Me button, tighten layout
│   │   ├─ FeaturedWork.js    # ← replace category cards with real projects
│   │   ├─ Contact.js         # ← add id="contact" for anchor
│   │   └─ Stats.js           # ← change "100% Passion" stat
│   ├─ pages/
│   │   ├─ index.js           # ← imports Hero + FeaturedWork
│   │   ├─ apps.js            # ← add search + filter
│   │   ├─ about.js           # ← rewrite bio, add "Currently Exploring"
│   │   ├─ stories.js         # ← fix Genesis Protocols date
│   │   └─ 404.js
│   ├─ templates/
│   │   └─ story-template.js   # story detail pages
│   ├─ styles/
│   │   └─ global.css         # add theme toggle styles
│   └─ data/
│       └─ projects.js        # project metadata array

```

```

├─ static/
│   └─ apps/
│       ├── neural_theory_lab_v2.html    # ← add back-banner markup
│       ├── understanding-layer-demo.html
│       ├── edge-ai-sandbox.html
│       └─ ... (all standalone apps)
│   └─ Dennis-Carroll-Resume.pdf        # ← add resume PDF
├─ gatsby-config.js
└─ gatsby-browser.js

```

Component Reference

Theme Toggle (simplified, single toggle)

Prop	Type	Default	Description
size	number	20	Icon width/height in px
className	string	""	Additional CSS classes

StandaloneAppShell

Prop	Type	Required	Description
appTitle	string	yes	Displayed in the sticky bar
children	ReactNode	yes	The app content

FeaturedWork (new section)

Prop	Type	Default	Description
projects	FeaturedProject[]	hardcoded array	Override default 3 projects

```

interface FeaturedProject {
  title: string;
  description: string;
  tags: string[];
  image: string;           // path to screenshot
  links: { label: string; url: string }[];
  accent: 'cyan' | 'purple' | 'emerald' | 'amber';
  isNew?: boolean;
}

```

Implementation Order Recommendation

Day 1 — P0 + P1 (2–3 hours)

- 1. Fix Contact Me button
- 2. Fix footer dead links
- 3. Fix Genesis Protocols date
- 4. Simplify theme toggle
- 5. Replace 100% Passion stat
- 6. Add back-banner to 3 most-visited standalone apps (Neural Lab, Agent Trace, Edge AI)

Day 2 — P2 Content (4–6 hours) 7. Rewrite About page bio 8. Redesign FeaturedWork section with 3 real projects 9. Add color-gradient thumbnails to app cards 10. Add “Currently Exploring” section to About 11. Improve hero (add scroll indicator + tighten spacing)

Week 2 — P3 Features (when you have time) 12. Add search + filter to Apps page 13. Add resume download + availability badge 14. Particle animation performance fix 15. Add real screenshots to all project cards (ongoing)

Estimated File Changes Summary

File	Lines Changed	Priority
src/sections/Hero.js	+5, -2	P0
src/components/Footer.js	+2, -4	P0
src/pages/stories.js or data file	+1, -1	P0
src/components/ThemeToggle.js	rewrite	P1
src/pages/about.js	+20, -5	P1, P2
src/sections/FeaturedWork.js	rewrite	P2
src/pages/apps.js	+60, -5	P2, P3
src/components/ParticleBackground.js	+15	P3
static/apps/*.html	+30 per file	P1
src/components/StandaloneAppShell.js	new file	P1
src/pages/apps/neural-theory-lab.js	new file (optional)	P3